

SI 427 Quiz 2
Department of Mathematics, IIT Bombay

Saturday

11:00 - 12:00

October 25, 2025

15 marks

You must justify all your answers. Answers without justification may not be awarded any marks. You must explicitly verify that all the hypotheses of any standard theorems are satisfied in the context in which you use them.

Question 1.

1. (2 marks) Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Explain why

(a) $|X|$ and

(b)

$$X^+ = \begin{cases} X & \text{if } X \geq 0, \text{ and} \\ 0 & \text{otherwise,} \end{cases}$$

are random variables on Ω .

Solution: If $X : \Omega \rightarrow \mathbb{R}$ is a random variable and $g : \mathbb{R} \rightarrow \mathbb{R}$ is a measurable function, then $g(X)$ is a random variable on Ω . In particular, this holds when g is a continuous function.

The function $x \rightarrow |x|$ is continuous. Hence, $|X|$ is a random variable on Ω . (1 mark)

The function $g : \mathbb{R} \rightarrow \mathbb{R}$ given by $g(x) = x$ if $x \geq 0$, and 0 otherwise, is continuous. Then $X^+ = g(X)$, so X^+ is a random variable on Ω . (1 mark)

Alternatively, observe that $X^+ = \frac{|X|+X}{2}$, and since the sum of measurable functions is measurable, it follows that X^+ is measurable.

2. (3 marks) Let $(\Omega, \mathcal{M}, P)$ be a probability measure space, and let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. Suppose that $E[|X|^t]$ exists for some $t > 0$. Show that $E[|X|^s]$ exists for every s such that $0 < s < t$.

Solution: Let $A = \{\omega, |X(\omega)| \leq 1\}$. Note that $|X(\omega)|^s \leq |X(\omega)|^t$ for all $\omega \in A$. (1 mark)

We have

$$\begin{aligned} \int_{\Omega} |X|^s dP &= \int_A |X|^s dP + \int_{A'} |X|^s dP \\ &\leq \int_A dP + \int_{A'} |X|^t dP \leq P(A) + E[|X|^t] < \infty. \end{aligned}$$

(2 marks)

(1 mark for estimating each of the integrals correctly.)

Question 2.

1. (2 marks) Consider the probability measure space $([0, 1], \mathcal{B}, m)$, where $\mathcal{B}$ is the Borel σ -algebra on $[0, 1]$ and m is the Lebesgue measure. Give an example of a random variable $X : [0, 1] \rightarrow \mathbb{R}$ for which $E[X]$ exists but $E[X^2]$ does not exist.

Solution: Let X be the function given by $X(x) = \frac{1}{\sqrt{x}}$ if $x \neq 0$, and $X(0) = 0$. Then

$$E[|X|] = E[X] = \int_0^1 \frac{dx}{\sqrt{x}} = 2\sqrt{x} \Big|_0^1 = 2.$$

However, $E[X^2] = \int_0^1 \frac{dx}{x}$ which does not exist. (2 marks)

2. (3 marks) Let $X = (X_1, X_2)$ be a random variable with joint PDF

$$f(x_1, x_2) = \begin{cases} \frac{1}{\Gamma(\alpha)\Gamma(\gamma)} x_1^{\alpha-1} (x_2 - x_1)^{\gamma-1} e^{-x_2} & \text{if } 0 < x_1 < x_2, \text{ and} \\ 0 & \text{otherwise.} \end{cases}$$

for $\alpha, \gamma > 0$.

- (a) (2 marks) Find the two marginal PDFs.

Solution: We have

$$\begin{aligned} f_1(x_1) &= \frac{1}{\Gamma(\alpha)\Gamma(\gamma)} x_1^{\alpha-1} \int_{x_1}^{\infty} (x_2 - x_1)^{\gamma-1} e^{-x_2} dx_2 = \frac{1}{\Gamma(\alpha)\Gamma(\gamma)} x_1^{\alpha-1} \int_0^{\infty} u^{\gamma-1} e^{-(u+x_1)} du \\ &= \frac{1}{\Gamma(\alpha)\Gamma(\gamma)} x_1^{\alpha-1} e^{-x_1} \Gamma(\gamma) = \frac{1}{\Gamma(\alpha)} x_1^{\alpha-1} e^{-x_1}. \end{aligned} \quad (1 \text{ mark})$$

Similarly,

$$\begin{aligned} f_2(x_2) &= \frac{1}{\Gamma(\alpha)\Gamma(\gamma)} e^{-x_2} \int_0^{x_2} x_1^{\alpha-1} (x_2 - x_1)^{\gamma-1} dx_1 = \frac{1}{\Gamma(\alpha)\Gamma(\gamma)} e^{-x_2} \int_0^{x_2} x_1^{\alpha-1} (x_2 - x_1)^{\gamma-1} dx_1 \\ &= \frac{1}{\Gamma(\alpha)\Gamma(\gamma)} e^{-x_2} x_2^{\gamma+\alpha-2} \int_0^1 u^{\alpha-1} (1-u)^{\gamma-1} du \\ &= \frac{1}{\Gamma(\alpha)\Gamma(\gamma)} e^{-x_2} x_2^{\gamma+\alpha-1} B(\alpha, \gamma) = \frac{1}{\Gamma(\alpha+\gamma)} x_2^{\gamma+\alpha-1} e^{-x_2}. \end{aligned} \quad (1 \text{ mark})$$

- (b) (1 mark) Find $F_{X_2|X_1}(x_2|x_1)$.

Solution: We have

$$f_{X_2|X_1}(x_2|x_1) = \frac{f(x_1, x_2)}{f_1(x_1)} = \frac{1}{\Gamma(\gamma)} e^{x_1} (x_2 - x_1)^{\gamma-1} e^{-x_2}.$$

for the conditional PDF of X_2 given $X_1 = x_1$. (1 mark)

Hence,

$$F_{X_2|X_1}(x_2|x_1) = \int_{-\infty}^{x_2} \frac{f(x_1, v)}{f_1(x_1)} dv = \frac{1}{\Gamma(\gamma)} e^{x_1} \int_{-\infty}^{x_2} (v - x_1)^{\gamma-1} e^{-v} dv.$$

As discussed in the tutorial, there is no closed form for this integral so 1 mark was awarded if the conditional PDF was written correctly.

Question 3.

- (2 marks) Let X be the IQ of a randomly selected person in a population. We are given that $E[X] = 100$ and $\text{var}(X) = 15$. Estimate the probability that a randomly chosen person has an IQ greater than or equal to 145.

Solution: Recall that if $E[X] = \mu$ and $E[X^2]$ both exist, we can use Chebyshev's inequality which says for any $K > 0$,

$$P_0 = P(\{\omega \mid |X(\omega) - \mu| \geq K\sigma\}) \leq \frac{1}{K^2}$$

where $\sigma^2 = \text{Var}(X)$. We are given $\mu = 100$ and $\sigma^2 = 15$. We would like $K\sigma = 45$, so we take $K = 3\sqrt{15}$. It follows that $P_0 < 1/135$.

Alternatively, we can write Chebyshev's inequality as

$$P_0 = P(\{\omega \mid |X(\omega) - \mu| \geq \delta\}) \leq \frac{\text{Var}(X)}{\delta^2} = \frac{15}{45^2} = \frac{1}{135},$$

if we take $\delta = 45$.

- Let $X_1, X_2 \sim \mathcal{N}(0, 1)$ be independent random variables. Let $Y_1 = \frac{X_1 + X_2}{\sqrt{2}}$ and $Y_2 = \frac{X_1 - X_2}{\sqrt{2}}$.

(a) (2 marks) What is the joint PDF of Y_1 and Y_2 ?

(b) (1 mark) Are Y_1 and Y_2 independent?

Solution: The PDFs of X_1 and X_2 are

$$f_1(x_1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x_1^2}{2}} \quad \text{and} \quad f_2(x_2) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x_2^2}{2}}$$

respectively. Since X_1 and X_2 are independent, the joint distribution of $X = (X_1, X_2)$ is given by

$$f_1(x_1)f_2(x_2) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x_1^2}{2}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x_2^2}{2}}.$$

Let $g(x_1, x_2) = \left(\frac{x_1 + x_2}{\sqrt{2}}, \frac{x_1 - x_2}{\sqrt{2}} \right)$, so $Y = (Y_1, Y_2) = g(X)$. Obviously, $h = g^{-1} = g$, so

$$h(y_1, y_2) = \left(\frac{y_1 + y_2}{\sqrt{2}}, \frac{y_1 - y_2}{\sqrt{2}} \right).$$

We have

$$Jh(y_1, y_2) = \begin{vmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{vmatrix} = -1$$

Hence $|Jh(y_1, y_2)| = 1$. It follows that the joint distribution of Y is given by

$$f_Y(y_1, y_2) = f(y_1, y_2) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y_1^2}{2}} \frac{1}{\sqrt{2\pi}} e^{-\frac{y_2^2}{2}}.$$

Clearly Y_1 and Y_2 are independent since

$$f_Y(y_1, y_2) = f_1(y_1)f_2(y_2).$$